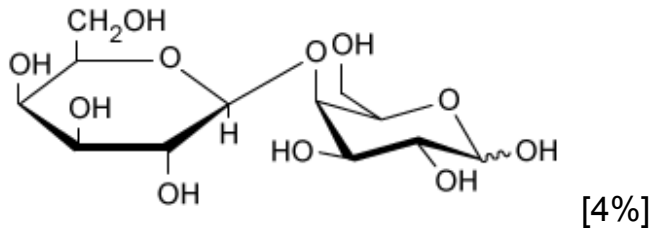
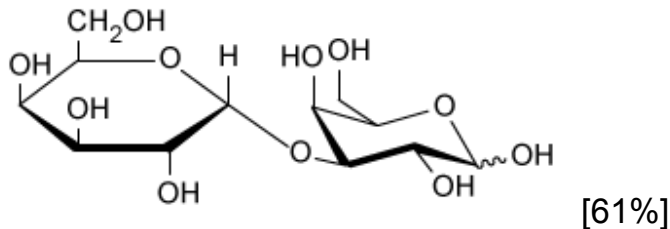


Alpha-gal syndrome is an allergic reaction to red meat that develops after being bitten by the lone star tick. The tick directly injects the Gal-($\alpha 1 \rightarrow 3$)-Gal disaccharide (also called α -Gal) into the bloodstream, triggering an allergic response to future oral consumption of α -Gal, which is present in all non-primate mammals. Which of the following Gal-Gal disaccharides represents the structure of α -Gal?

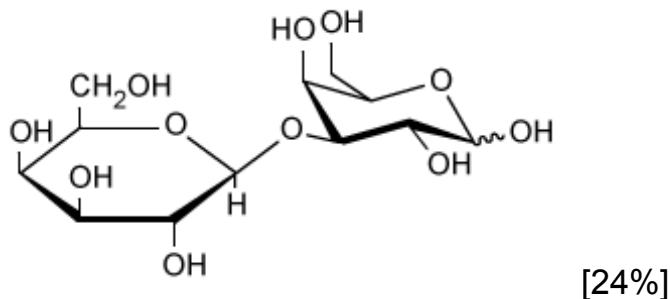
☐ A.



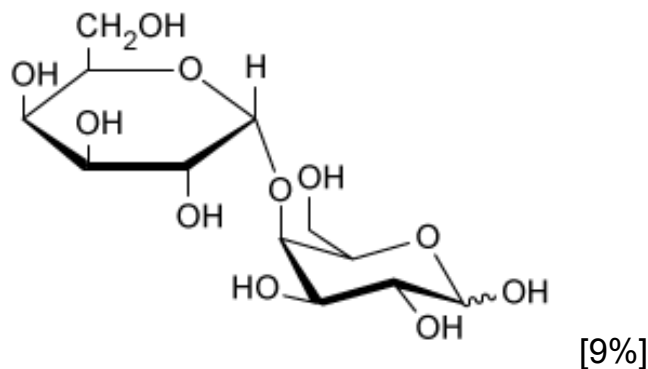
✓ ☒ B.



☐ C.



☐ D.

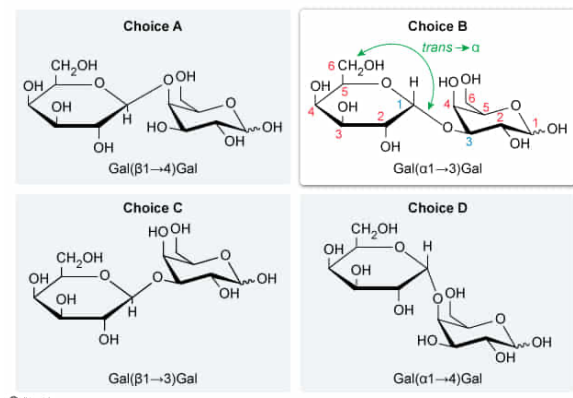


Correct

62% answered correctly

Explanation:

Galactose(α 1 $\rightarrow$ 3)Galactose



Carbohydrates can be joined to other molecules, including other carbohydrates, by **glycosidic bonds**. These bonds involve the **anomeric carbon** of the carbohydrate and a nucleophilic functional group (eg, hydroxyl group) of the second molecule. Different disaccharides composed of the same two precursor monosaccharides can be distinguished from each other by the **configuration** of the anomeric carbon and the **positions** of the carbons involved in the glycosidic bond connecting the monosaccharides.

The question states that the two **galactose** monosaccharides are connected by an **α 1 $\rightarrow$ 3 glycosidic bond**. This means that the anomeric carbon (**C1**) of one galactose is in the α -configuration and connected to the **C3** carbon of another galactose through a glycosidic bond. In each available choice, the monosaccharide on the left is contributing **its anomeric carbon** to the glycosidic bond. Therefore, the monosaccharide on the right must be involved in the glycosidic bond through the hydroxyl group on **its C3 carbon**.

The anomeric carbon of a cyclic monosaccharide can adopt **one of two configurations**, depending on the stereochemistry of the **anomeric hydroxyl**. In cyclic projections, the **α -configuration** displays the **anomeric hydroxyl *trans* to (on the opposite side of) the C6 hydroxymethyl group with respect to the ring** whereas the β -configuration displays the anomeric hydroxyl *cis* to (on the same side of) the C6 hydroxymethyl. Nomenclature of anomeric carbons involved in glycosidic bonds follow similar conventions.

Because the question states the glycosidic bond is an **α 1 $\rightarrow$ 3 glycosidic bond**, the anomeric carbon involved in the glycosidic bond must have the glycosidic bond *trans* to the monosaccharides C6 hydroxymethyl, as in the structure shown in Choice B.

(Choices A and D) These structures show glycosidic bonds connected to C4 of the second monosaccharide, and would be 1 $\rightarrow$ 4 glycosidic bonds, not 1 $\rightarrow$ 3 glycosidic bonds.

(Choice C) This structure shows a 1 $\rightarrow$ 3 glycosidic bond; however, the glycosidic bond to the anomeric carbon is *cis* to the C6 hydroxymethyl. Therefore, this is a β 1 $\rightarrow$ 3 glycosidic bond, not an α 1 $\rightarrow$ 3 glycosidic bond.

Educational objective:

Glycosidic bonds are linkages between the anomeric carbon of a carbohydrate and a

nucleophilic functional group of a second molecule. In cyclic hexoses, α -glycosidic bonds are *trans* to the C6-hydroxymethyl and β -glycosidic bonds are *cis* to the C6 hydroxymethyl.

Time spent:0 sec

Subject:
Biochemistry

QID:403838

System:
Carbohydrates, Nucleotides, and Lipids